

Q. Can someone please suggest me a good book for learning PIC MCUs' architecture and its interfacing with different peripherals like LEDs, LCD and keypad? Also, in which cases is Microchip's PIC better than Atmel's AVR?

A. There are various authors writing on this topic as PIC gains popularity among industrial product designers as well as hobbyists. The selection of a right book depends on two things: prerequisites and scope.

You must carefully check the prerequisites section of the book to understand what is expected from you before you start. If you are a beginner and the book requires you to have some prior knowledge of microcontrollers and 'C' language, it is not for you.

Checking the scope of the book is equally important in judging a book. The book should have everything you are looking for.

To begin with PIC microcontrollers, I would recommend 'PIC Microcontroller and Embedded Systems: Using Assembly and C' by Muhammad Ali

Mazidi, Rolin McKinlay and Danny Causey. This book starts with the very basic concepts and covers both the Assembly and C languages. It also covers well the interfacing with peripherals such as keyboard matrix, LCD, motors and serial ports.

For in-depth study of PIC MCUs, you may refer to 'Fundamentals of Microcontrollers and Applications in Embedded Systems' by Ramesh S. Gaonkar. This book explains the microcontroller fundamentals and applications in embedded systems very nicely.

To answer your question on PIC and AVR MCUs, I would like to first mention that there is nothing like 'the best' in this field. You have to decide which microcontroller would be more appropriate for your application. This is a very frequently asked question and the discussion always becomes heated when someone tries to declare one better than the other.

In some respects, however, I find PIC better than AVR:

Small instruction set. PIC has a relatively small instruction set that varies from 35 to 80 instructions for different variants.

These instructions are simple and can be learnt very quickly. In AVR, the instruction set is not as simple as in PIC and also some things don't seem logical, like immediate XOR instruction doesn't exist.

Power. The power consumption of a PIC

MCU at 5V is much less than of AVR. But Atmel has recently launched a technology called *picoPower* that claims to reduce the power consumption while maintaining the performance. It uses techniques like sleepwalking, faster wakeup, multiple clock domains, DMA controller and event system, and true 1.62V operation to reduce the overall power consumption.

ADC. PIC has a 10-bit, multi-channel analogue-to-digital converter which is faster than AVR.

Interrupts. In AVR you will always get some jitters (refer Fig. 1), whereas in PIC latency is always consistent.

Clocks. The R-C oscillator of the PIC is more accurately calibrated.

Input/output (I/O). The drive capability of I/O ports is much better than in AVR. The ports are more rugged for driving the devices directly. Some extra milliamps do not cause voltage drop.

Programming. Programming PIC is easy and the same platform is compatible with all the parts. In AVR, there are several AVR ICE variants for different parts.

Availability. Availability of the components in the market depends on various things like minimum-order quantities (MOQs), lead times and obsolescence:

MOQ. Atmel has very strict MOQ and the authorised distributors are not allowed to open the sealed package. On the other hand, you can easily get small quantities from Microchip.

Lead time. The lead time with Atmel—six weeks on an average—is also more than with Microchip.

Obsolescence. Microchip always keeps old parts available long after they have been superseded. Atmel, on the other hand, has a higher obsolescence rate and immediately pulls the chips from the market.

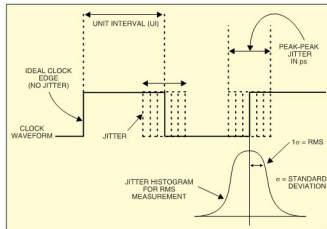


Fig. 1: Jitter



Fig. 2: ICE (Image courtesy: www.microchip.com)

Answer compiled by EFY technical editor, Ankit Gupta.

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